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The Discoverable Solution Set of Planar Path Synthesis: A Multimodal Capability Benchmark of Eight Mechanism Families under Explicit Packaging Limits

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Abstract

Planar path synthesis requires selecting a mechanism architecture and dimensions that meet accuracy requirements within geometric limits. We compare eight planar mechanism families using a common search engine and physical errors on synthetic closed curves normalized to 90 mm width. Campaigns use 30 seeds; root mean square (RMS) success requires an error of at most 1 mm, while strict success additionally requires a maximum error of at most 2 mm. In the initial medium-target comparison under compact bounds, Watt I achieves RMS success in 25 of 30 seeds, Stephenson III in two and the four-bar in none (nominal Freeman–Halton $p = 5.48 \times 10^{-15}$). A separate all-family campaign yields strict designs in three families. On the medium-hard target, tightening link-length bounds from 260 to 180 mm raises Watt I median RMS error from 0.414 to 0.721 mm (Wilcoxon $p = 0.015$), without a significant pass-rate change. A quality-diversity extension redistributes strict discoveries across families at approximately matched cost, although differences are not significant after Holm correction. These results support comparing architectures, geometric constraints and search policies when selecting candidate mechanism designs.

Keywords: mechanism synthesis; computational mechanical design; path generation; planar linkages; packaging constraints; comparative optimization; design-space exploration; quality diversity

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1. Introduction

Selecting a planar mechanism for a specified path requires choosing an architecture and dimensions that satisfy accuracy requirements within allowable geometric bounds. This is a computational mechanical design problem: different architectures may offer feasible alternatives, but a single optimization run need not reveal them. Dimensional synthesis is normally posed inside a chosen topology: a four-bar, a slider-crank or a specific six-bar is assumed, and the work optimizes its dimensions [1–11]. A complementary design question is therefore important. Given a target curve and a packaging envelope, which families can meet a stated tolerance at all, and how reliably?

That silence matters because the two questions have different answers. A sub-millimeter result for one linkage under one budget says little about whether another family could have done better, or whether the first would succeed again from a different start. It also determines what may be concluded from failure: if a family fails under a

bounded search, the defensible statement is that it did not demonstrate capability under the stated parameterization, constraints and budget — not that it is incapable.

The inverse problem is, in addition, multimodal. Population-based and evolutionary metaheuristics are widely applied to mechanism path synthesis, from particle swarm optimization [12] and differential evolution [13] to krill herd algorithms [14,15] and teaching–learning optimization [16,17], with comparative studies and new algorithmic variants still appearing [18,19]; problem-specific error formulations such as the circular proximity function [20] and Fourier-based descriptors [21] complement them. Applied to a prescribed target, these methods can recover different dimension sets for the same task [22–27], a multiplicity already present in classical numerical and complete-solution studies of four-bar synthesis [28,29]. Six-bar synthesis studies further show that successful designs can be obtained across several topologies and task formulations [30–36], and recent learning-based and graph-based approaches have further expanded multi-solution and multi-architecture linkage synthesis [37,38], while uncertainty-aware synthesis adds another layer of variability to single-design claims [39]. What is missing is a common-engine, common-metric comparison of how reliably each family can be *found* to meet a tolerance, reported as rates over independent searches rather than as best cases.

The contributions of this work are as follows.

1. We cast family-level path synthesis as a multimodal inverse problem and use seed-wise success rates under one common search engine as a reproducible measure of the non-uniqueness that is actually discoverable, in contrast to the best-case designs reported in most synthesis studies.
2. We provide a paired 2×2 packaging experiment on shared seeds that isolates the measurable cost of an explicit packaging limit on accuracy and on pass rates.
3. We show, with a budget-matched quality-diversity extension built on MAP-Elites and quality-diversity methods [40–42], that the family composition of the strict-solution set is conditional on the search policy, so that single-policy capability rankings — including ours — must be read as search-dependent.
4. We report negative and null results explicitly, with paired tests and Holm-adjusted inference [43], so that non-significant differences are not mistaken for equivalence.

This paper reports a capability benchmark for planar closed-curve generation in this framing. Eight families are searched by one common engine, evaluated by one physical error metric in millimeters, and compared by seed-wise success rates rather than by a single best design. Three results follow. The families separate sharply on a medium-difficulty target under an explicit link limit. Running a harder target under two packaging envelopes, with nothing else changed, isolates what the limit costs. And replacing only the outer search loop, at a comparable simulation budget, changes which family supplies the designs that meet the stricter tolerance — so a capability map is conditional on the search that produced it.

2. Materials and Methods

2.1. Families, Targets and Packaging Profiles

Eight planar families are considered: four-bar, slider-crank, Watt I, Watt II, Stephenson I, Stephenson II, Stephenson III and geared five-bar. Each is parameterized by link lengths, ground pivots, tracer offsets and family-specific quantities; Figure 1 shows their kinematic graphs. Every candidate is clipped into the search bounds of its family before it is simulated, so no evaluated candidate lies outside them. A candidate is then rejected when its kinematic solver reports an invalid configuration or fails to produce a path. Geometric feasibility is a precondition: a candidate for which the kinematic solver fails is never scored on accuracy.

Proximity to singularity is also discouraged through a penalty in the composite search cost; it is not represented here as a universal hard exclusion.

The packaging profile acts through those bounds. The compact profile limits coupler and secondary links to 180 mm; the relaxed profile raises them to 260 mm, with ground links, cranks, tracer radii and offsets scaled correspondingly. Nothing else changes between profiles: the simulator, the error metric, the acceptance tolerances and the refinement stage are identical. One asymmetry is worth stating: the slider-crank connecting rod is bounded at 210 mm against the 180 mm compact limit, the only link bound that exceeds it. Thus, packaging here means family-specific parameter bounds, rather than a verified global occupied volume or a collision-free installation envelope.

Three synthetic closed targets are used — easy, medium and medium-hard — each normalized to a width of 90 mm before synthesis and before any metric is computed. The recorded benchmark configurations use $N = 120$ points. The archived target paths and per-run configurations identify the target and bounds of each campaign; the initial medium-family comparison and the later all-family campaign are separate experiments, not interchangeable samples.

2.2. Error Metric and Acceptance

Candidate and target curves are resampled by arc length and aligned by cyclic phase shift and rigid-body transformation with no scaling [44–46], so the reported error is a physical distance in millimeters on a 90 mm curve. Formally, with arc-length-resampled candidate and target points $\{p_i\}_{i=0}^{N-1}$ and $\{t_i\}_{i=0}^{N-1}$, the alignment is the joint minimization over the cyclic shift k, a rotation $R \in \mathrm{SO}(2)$ and a translation $c \in \mathbb{R}^2$; no scaling is applied:

$$E_{\mathrm{RMS}} = \min_{k,R,c} \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} \left\| R\,p_i + c - t_{(i+k) \bmod N} \right\|^2}, \tag{1}$$

and $E_{\max}$ is the largest pointwise distance $\|R\,p_i + c - t_{(i+k) \bmod N}\|$ under the same minimizers, so both errors refer to one and the same physical alignment. Two criteria are used. The root mean square (RMS) criterion is $E_{\mathrm{RMS}} \leq 1$ mm. The strict criterion additionally requires the pointwise maximum,

$$E_{\mathrm{RMS}} \leq 1\ \mathrm{mm} \quad \text{and} \quad E_{\max} \leq 2\ \mathrm{mm}, \tag{2}$$

which is the criterion that matters for applications requiring pointwise accuracy.

2.3. Search Engine and Statistical Design

One common engine searches every family: random sampling within bounds, shape-seeded templates, selection by a composite cost dominated by RMS error, and a least-squares refinement stage. Table 1 reports the budget of each run group. Capability is reported as a seed-wise rate over 30 independent runs, rather than as a best case. Recorded benchmark configurations identify Python 3.11.15 and NumPy 2.4.6. The released synthesis scripts define the family parameter order and bounds, sampling, objective weights and refinement settings; the archived configurations and target paths are needed together to reproduce a campaign.

A supplementary sensitivity analysis treats shared seed identifiers as blocks. It uses an exact within-seed label permutation for the initial three-family medium comparison, exact McNemar tests for paired family success indicators and Holm adjustment within the seven all-family contrasts and the four policy contrasts. This reanalysis uses archived outputs and does not replace the numerical contrasts reported in the main tables.

Anthropic Claude Opus assisted in constructing the analysis and experiment code, as well as language editing. OpenAI Codex assisted with the editorial adaptation and audit scripts used to cross-check archived outputs; no new synthesis experiments were introduced in this revision.

The easy target screens all eight families. The medium target under compact geometry is the family-discriminating experiment. The medium-hard target is evaluated for Watt I and the geared five-bar under both packaging profiles, using the same 30 seeds within each family, which makes the within-family envelope comparison paired.

Table 1. Search-budget parameters used in the statistical runs. Easy: all eight families. Medium: the compact-geometry comparison. Medium-hard: both families under both packaging profiles.

Run group	Easy	Medium	Medium-hard
Random samples	48	120	900
Shape seeds	10	16	36
Refined candidates	2	4	8
Elite neighborhoods	5	6	8
Neighbor samples	4	4	6
DE restarts	0	0	2
Local iterations	55	100	220
LS evaluations	140	300	800

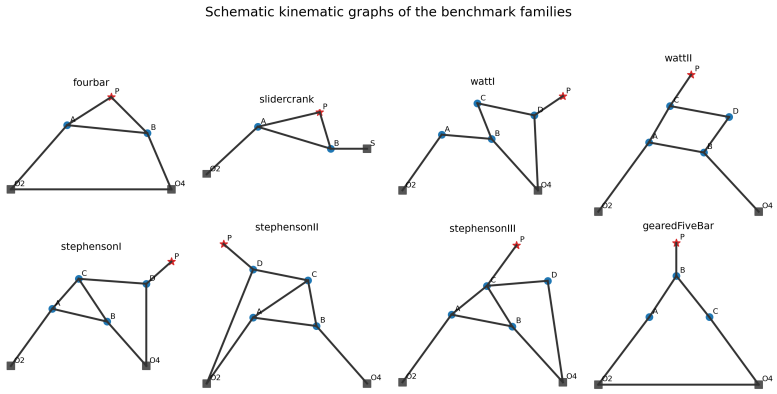


Figure 1. Schematic kinematic graphs of the eight benchmark families. The drawings are topological schematics used to identify the compared families; dimensions are optimized separately during synthesis.

3. Results

3.1. The Easy Target Does Not Discriminate

The easy all-family experiment succeeds in all 30 seeds with a median winning RMS error of 0.022 mm, two orders of magnitude below the criterion. That value is itself the finding: a benign target cannot separate families, because even simple ones reproduce it comfortably. Watt I, Stephenson III, four-bar, Stephenson I and slider-crank pass in 30 of 30 seeds and the geared five-bar in 26 of 30. Watt II passes in one of 30 seeds, whereas Stephenson II passes in none. Seven families therefore have at least one successful design, but repeatability differs substantially. Rare success and bounded-search failure should not be interpreted as kinematic impossibility.

3.2. The Medium Target Separates the Families

In the initial three-family medium-target campaign, under compact geometry and an identical per-family budget, the families separate sharply (Table 2). Watt I passes the RMS criterion in 25 of 30 seeds with median 0.850 mm, Stephenson III in 2 of 30 with median

1.203 mm, and the four-bar in none with median 1.395 mm. The nominal independent-sample contrast gives $p = 5.48 \times 10^{-15}$ by the Freeman–Halton extension of the Fisher exact test [47].

The strict criterion is much harder than the RMS one. Watt I meets it in 7 of 30 seeds against 25 of 30 for the RMS criterion alone, and that gap is partly a property of an RMS-dominated objective rather than of the families: the search minimizes a composite cost dominated by RMS error, so the strict pass rate measures the strict-tolerance capability of an RMS-oriented pipeline.

A separate campaign applying the same policy to all eight families on this target, over 30 seeds, shows that strict designs are not unique to Watt I (Table 3). The geared five-bar reaches the strict criterion in 6 of 30 seeds and Watt II in one, whose design was re-simulated in an independent process and reproduces its reported errors of 0.649 mm RMS and 1.528 mm maximum. Three families admit at least one design meeting the strict tolerance in this all-family campaign: Watt I, the geared five-bar and Watt II. A single successful seed establishes an observed feasible design, not reliable recovery across runs. Watt I reaches RMS success in 27 of 30 seeds and strict success in 16, compared with 25 and 7 in the separate initial campaign.

Table 2. Family-level capability statistics over 30 random seeds per row. Root mean square (RMS) pass denotes runs with RMS error at most 1 mm; strict pass also requires maximum error at most 2 mm. Pass counts are out of 30, so the percentages are omitted. Median, IQR and range refer to RMS error; IQR denotes interquartile range. All errors are in millimeters.

Target	Family	RMS pass	Strict pass	Median	IQR	Range
easy	Watt I	30/30	30/30	0.022	[0.015, 0.026]	0.013–0.065
easy	Stephenson III	30/30	30/30	0.116	[0.088, 0.149]	0.039–0.252
easy	geared five-bar	26/30	26/30	0.190	[0.116, 0.485]	0.026–1.142
easy	four-bar	30/30	30/30	0.199	[0.131, 0.258]	0.061–0.371
easy	Stephenson I	30/30	30/30	0.304	[0.210, 0.406]	0.061–0.707
easy	slider-crank	30/30	30/30	0.667	[0.608, 0.690]	0.254–0.711
easy	Stephenson II	0/30	0/30	19.005	[19.002, 19.009]	18.955–19.075
easy	Watt II	1/30	1/30	19.009	[18.999, 19.009]	0.058–19.067
medium	Watt I	25/30	7/30	0.850	[0.628, 0.952]	0.248–1.200
medium	Stephenson III	2/30	0/30	1.203	[1.139, 1.256]	0.998–1.374
medium	four-bar	0/30	0/30	1.395	[1.324, 1.400]	1.277–1.409
medium-hard	Watt I	25/30	25/30	0.414	[0.268, 0.625]	0.236–2.025

Table 3. Medium target, compact geometry, all eight families under identical conditions and search configuration, 30 seeds per family. Pass = RMS $\leq$ 1 mm; strict = RMS $\leq$ 1 mm and maximum error $\leq$ 2 mm. CI is the Wilson 95% interval [48] for the pass rate. p_{Holm} is the nominal two-sided Fisher exact test of each family's pass count against Watt I, Holm-corrected within this family of seven comparisons.

Family	Pass	Strict	Pass rate CI	Median RMS	IQR	p_{Holm} vs Watt I
Watt I	27/30	16/30	[0.74, 0.97]	0.786	[0.432, 0.925]	—
Stephenson III	0/30	0/30	[0.00, 0.11]	1.235	[1.199, 1.282]	<0.001
Stephenson I	0/30	0/30	[0.00, 0.11]	1.394	[1.352, 1.403]	<0.001
Geared five-bar	9/30	6/30	[0.17, 0.48]	1.110	[0.982, 1.252]	<0.001
Slider-crank	0/30	0/30	[0.00, 0.11]	1.831	[1.761, 1.862]	<0.001
Four-bar	0/30	0/30	[0.00, 0.11]	1.395	[1.324, 1.400]	<0.001
Watt II	1/30	1/30	[0.01, 0.17]	25.219	[12.295, 25.265]	<0.001
Stephenson II	0/30	0/30	[0.00, 0.11]	25.246	[8.362, 25.267]	<0.001

3.3. The Packaging Envelope

The medium-hard target was run for Watt I and the geared five-bar under both envelopes, on the same seeds and with every other setting held fixed, giving a two-family by two-envelope design with 30 seeds per cell (Table 4). Because the envelopes share their seeds, the within-family comparison is paired and is tested as such: exact McNemar [49]

for the paired rates, computed from the discordant pairs using the exact binomial test implemented in SciPy [50], and the Wilcoxon signed-rank test [51] for the error.

Within this 30-seed experiment, tightening the envelope shifted Watt I RMS errors upward in the paired Wilcoxon comparison. The median rose from 0.414 mm to 0.721 mm ($p = 0.015$), with the worst seed rising from 2.025 mm to 4.359 mm, while no statistically significant change was detected in either pass rate ($p = 0.30$ and $p = 0.21$). Absence of a detected difference at this sample size is not evidence of equivalence. For the geared five-bar no envelope effect was detected in either the rates, 4 of 30 against 3 of 30, or the median error, 1.402 mm against 1.261 mm. The reported nominal between-family Fisher contrasts are large under both envelopes, at $p = 7.0 \times 10^{-8}$ relaxed and $p = 3.3 \times 10^{-5}$ compact.

Within the two families tested, then, the packaging envelope measurably affected Watt I accuracy but produced no detectable change in geared five-bar performance. Whether that pattern holds more widely is not established by two families.

Table 4. Medium-hard target under two packaging envelopes and two families, 30 seeds per cell. The four campaigns share the target curve, the sampling budget, the differential-evolution stage and the refinement depth; their recorded configurations differ only in the family and in the two fields that define the envelope. Strict criterion: RMS ≤ 1 mm and maximum error ≤ 2 mm. The two envelopes were run on the same seeds 1–30, so the comparison within a family is paired: p is the exact McNemar test for the rates and the Wilcoxon signed-rank test for the error. Between families within an envelope the reported contrasts use the independent-sample Fisher exact formulation; the nominal strict-rate results are $p = 7.0 \times 10^{-8}$ (relaxed) and $p = 3.3 \times 10^{-5}$ (compact).

Outcome	Relaxed, 260 mm	Compact, 180 mm	p
Watt I			
Passes RMS ≤ 1 mm	25/30	20/30	0.30
Meets strict criterion	25/30	19/30	0.21
Median RMS error (mm)	0.414	0.721	0.015
Geared five-bar			
Passes RMS ≤ 1 mm	5/30	3/30	0.73
Meets strict criterion	4/30	3/30	1.00
Median RMS error (mm)	1.402	1.261	0.349

3.4. The Capability Map Depends on the Search Policy

The results above are conditional on one search policy. To test whether the strict-design map is invariant to it, the outer loop alone was replaced by a quality-diversity (QD) archive, holding the simulator, the evaluation function, the acceptance criteria and the refinement stage fixed. QD algorithms search for a diverse archive of high-quality solutions rather than a single optimum [40–42,52], and have since been extended to multi-objective [53] and uncertain [54] domains.

Candidates are maintained in a MAP-Elites archive [42] over two hand-defined, rotation-invariant curve descriptors — the elongation of the coupler curve and the logarithm of its turning energy — on a 24×24 grid, the conventional grid instantiation of an archive whose centroidal-Voronoi variant targets higher-dimensional descriptor spaces [55]. Admission and ranking are governed by different quantities: geometric feasibility is a precondition, and archive quality then ranks the feasible candidates competing for the same cell as the RMS path error alone,

$$Q(x) = -E_{\mathrm{RMS}}(x), \tag{3}$$

so $E_{\max}$ is retained and reported for every elite and enters the strict criterion but does not participate in the cell-level competition. All eight families compete in a single archive. Fifteen seeds were run for the baseline policy and fifteen for the extension at each of two

search budgets, 60,000 and 34,000 evaluations, the second chosen so that total simulations approximately match the baseline; every simulation, including refinement, is counted.

Table 5 reports the outcome. All three arms reach the strict criterion in about the same fraction of seeds: 9 of 15 for the baseline, 10 of 15 for each extension arm. What changes is which family supplies those designs. The baseline reaches the criterion through the geared five-bar in 1 seed of 15 and through Watt I in 9; the approximately budget-matched extension reaches it through the geared five-bar in 7 and through Watt I in 7. The set of strict-capable families is the same in both arms; what the extension changes at comparable cost is the frequency with which each is discovered.

The strength of that evidence should be stated carefully. The geared five-bar difference has an uncorrected Fisher $p = 0.035$, but the table makes four comparisons on the same data and after Holm adjustment none falls below 0.05. The redistribution is therefore a large descriptive effect reported as exploratory, not an established significant difference. The matched arm still spends 6.7% more simulations than the baseline, which substantially reduces, but does not eliminate, budget as an alternative explanation.

These experiments do not establish dominance of either policy. The most accurate single design of the whole experiment came from the budget-matched extension, at RMS 0.286 mm, against 0.303 mm for the baseline. Independent re-simulation of the 38 strict extension designs whose parameter vectors are included in the archived JSON files reproduces their errors to numerical precision. The baseline JSON summaries omit those vectors and do not, by themselves, support the same independent check.

Table 5. Search policies on the medium target under compact geometry, 15 seeds each, same simulator, same acceptance criteria and same least-squares refinement stage. The baseline draws 120 random samples and 16 shape-seeded templates per family with differential evolution disabled; the extension couples a MAP-Elites archive with surrogate-guided variation [56,57]. Strict criterion: RMS ≤ 1 mm and maximum error ≤ 2 mm. Intervals are Wilson 95% intervals [48] for the seed-wise rate of the approximately budget-matched extension. The raw p is the nominal two-sided Fisher exact test of that arm against the baseline; the Holm column adjusts the four comparisons of this table for multiplicity, and none of them remains below 0.05 after adjustment. The extension is reported twice: at its natural budget and at a search budget reduced from 60,000 to 34,000 evaluations so that total simulations come within 7% of the baseline. All simulation counts are measured.

		QD + surrogate			p	
Outcome	Baseline	natural	matched	95% CI (matched)	raw	Holm
Seeds with at least one strict design	9/15	10/15	10/15	[0.42, 0.85]	1.000	1.000
from geared five-bar	1/15	10/15	7/15	[0.25, 0.70]	0.035	0.141
from Watt I	9/15	1/15	7/15	[0.25, 0.70]	0.715	1.000
from Stephenson III	0/15	1/15	0/15	[0.00, 0.20]	1.000	1.000
Strict designs, total	12	19	19		—	—
Simulations, total	786,329	1,179,481	839,257		—	—
Best RMS of any seed (mm)	0.303	0.513	0.286		—	—

4. Discussion

Among the families tested, Watt I is the most repeatable on the discriminating target. The claim is conditional: Watt I is the most repeatable family *among those tested, under this envelope, with this search*. It is not a claim that Watt I is superior to all planar linkages, and Section 3.4 shows directly how the frequency and family composition of strict discoveries change under a different outer search policy. For a designer, the practical implication is that several families admit solutions within the same tolerance, and which of them a design process will actually surface is a property of the search as much as of the mechanism.

Raising the link limit from 180 mm to 260 mm changes the physical design envelope. It is not a numerical trick, but neither is it free, and running the medium-hard target under

both limits and both families makes its price measurable. Tightening the limit raised Watt I median error by about 0.3 mm and roughly doubled its worst case, while its strict-pass count fell from 25 of 30 to 19 of 30 by an amount this experiment cannot resolve. For the geared five-bar no envelope effect was detected at all. For a designer, these results motivate evaluating accuracy under the intended geometric bounds, rather than transferring a result obtained under relaxed bounds to a compact design. The experiment covers only two families and one target; its non-significant pass-rate contrasts do not establish equivalent reachability.

The realized Watt I design of Figure 2 shows that the reported errors correspond to a mechanism, not only to a number: its parameter vector is released, and re-simulating it reproduces its errors exactly.

4.1. Limitations

Several limitations bound these results. The published Fisher and Freeman–Halton contrasts use independent-sample contingency formulas. However, the all-family archive contains multiple families within each seed-wise run, and the policy arms share seed identifiers. Their between-family and between-policy p-values are retained as nominal contrasts; a common seed identifier does not by itself establish independence. The supplementary seed-blocked sensitivity analysis preserves the medium-target separation between families, while none of the four policy contrasts is significant after Holm correction. These checks support the qualitative interpretation without making the original independent-sample formulas paired tests. The QD redistribution remains exploratory after Holm correction. The within-family packaging comparisons explicitly use paired tests.

The targets are synthetic closed curves rather than industrial measured paths. The statistical protocol is not symmetric across all families at all difficulties: the medium-hard target covers two families, so the envelope effect is identified for that target but not across families, and no profile was run at both budgets, so packaging and budget remain confounded between the medium and medium-hard cases. The objective prioritizes RMS error, so additional local-error weighting is likely to be required for high-precision manufacturing. The families range from five design variables to twelve, so an identical number of evaluations does not represent an identical sampling density of the design space: the protocol equalizes effort, not effective search difficulty. Finally, every capability statement here is conditional on one search policy, and family rankings from single-policy benchmarks — including this one — should be read accordingly.

All results are from simulation; physical validation of a fabricated mechanism is left to future work, as is an extension in which a precomputed archive is used to recover performance after a geometric perturbation.

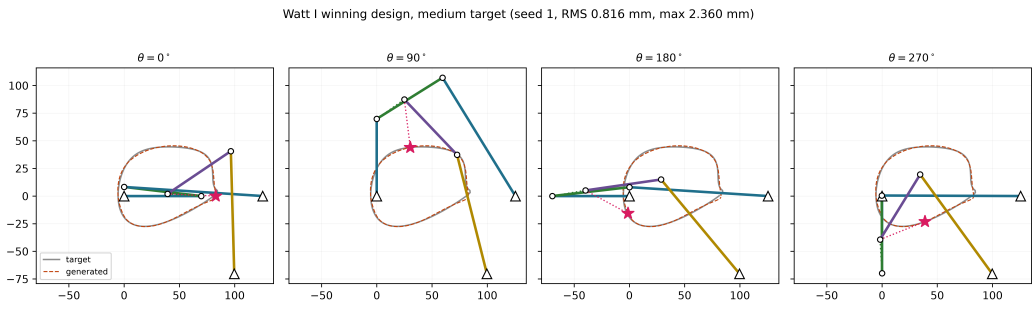


Figure 2. Watt I winning design for seed 1 on the medium target (RMS error 0.816 mm, maximum error 2.360 mm), shown at four crank positions. Triangles are ground pivots; circles are moving joints; the star is the tracer point. The grey curve is the target mapped into the mechanism frame by the inverse of the rigid alignment, so the overlay displays the actual fit.

5. Conclusions

The comparison supports considering several mechanism architectures and repeated searches when synthesizing a prescribed planar path. On the easy target, seven families have at least one RMS-successful design. On the medium target under compact geometry, Watt I is the most repeatable family under the tested policy; the separate all-family campaign finds strict designs in Watt I, the geared five-bar and Watt II. Tighter bounds increase Watt I RMS error in the paired medium-hard experiment, while the non-significant pass-rate contrasts do not establish equivalence. The quality-diversity comparison illustrates an exploratory redistribution of strict discoveries at approximately matched cost. Thus, the designs recovered for a task depend jointly on architecture, geometric constraints and the policy used to explore the design space. These conclusions concern the simulated targets and parameterizations tested here, with physical validation still pending.

Supplementary Materials: Supplementary code and data: synthesis engine, optional C++ geometry core, quality-diversity harness, archived campaign outputs, and audit and seed-blocked sensitivity-analysis scripts.

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Conflicts of Interest: The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

RMS	Root mean square
QD	Quality diversity
DE	Differential evolution
CI	Confidence interval

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